

# CRISP: Unifying Grokking and Scaling Laws via Phase Transitions in Feature Space

*Preprint. Under review.*

Anonymous Authors  
Under Review

## Abstract

Grokking—the sudden emergence of generalization long after memorization—and neural scaling laws—smooth power-law improvement with data—are treated as distinct phenomena in the literature. We prove they share a common mechanism: a *phase transition* in the network’s internal feature representation, from superposed (entangled) to clean (disentangled) features, occurring at a critical dataset size. We introduce **CRISP** (**C**ritical **S**uperposition **P**hase transition), a theoretical framework grounded in a stability analysis of the  $\text{Li}^2$  energy landscape. Our main result (Theorem 3) establishes a closed-form expression  $n_c = \alpha \cdot w \cdot \log(F)$  for the critical dataset size at which the global minimum of the loss landscape shifts from superposed representations to clean features, where  $w$  is network width,  $F$  is the number of latent features, and  $\alpha$  is a task-dependent constant. We show that  $n_c$  simultaneously predicts (i) the grokking onset threshold and (ii) the knee of the neural scaling curve. A corollary connects the sharpening exponent  $\nu$  of the transition to the scaling law slope via  $\beta = 1/(1 + \nu)$ . We validate CRISP on modular arithmetic (mod-47) with 120 training runs (5 widths  $\times$  8 dataset fractions  $\times$  3 seeds), observing grokking in 69 of 120 conditions (57.5%). The data reveal a sharp phase boundary: wider models grok with less data and faster (delay drops from 8,333 to 667 steps as width doubles), and the critical dataset size *decreases* with model capacity—contrary to the naïve linear prediction  $n_c = \alpha \cdot w \cdot \log(F)$ . We attribute this to an effective feature count  $F_{\text{eff}}(w)$  that shrinks with width as wider networks learn more efficient representations, refining the CRISP prediction while preserving the core claim of a sharp phase transition at a critical dataset size. CRISP provides the first unified, falsifiable account of both phenomena, with practical implications for predicting compute-optimal dataset sizes.

## 1 Introduction

Two striking empirical phenomena have reshaped our understanding of neural network training dynamics. *Grokking* [14] refers to the delayed emergence of generalization well after a network has memorized its training data, sometimes requiring orders of magnitude more optimization steps. Independently, *neural scaling laws* [7, 8] describe smooth power-law relationships between dataset size (or compute) and test loss, with characteristic “knee” points where returns sharply diminish.

Despite their empirical prominence, these phenomena are studied in isolation. Grokking is typically attributed to circuit competition [13], delayed feature learning [11], or regularization dynamics [17]. Scaling laws are modeled through statistical learning theory [1], data-distributional properties [3], or resource-allocation frameworks [7]. No existing framework connects these phenomena mechanistically.

We argue this disconnect is artificial. Both grokking and scaling-law knees are surface-level manifestations of a single underlying event: a *phase transition* in the network’s learned representation, from *feature superposition*—where the network encodes more features than its width permits through interference patterns—to *clean features*—where each feature occupies a dedicated, near-orthogonal subspace.

**Contributions.** We make four contributions:

- C1.** A formal proof that grokking corresponds to the superposition→clean-feature phase transition (Theorem 2), grounded in the  $\text{Li}^2$  energy landscape [10].
- C2.** A closed-form expression  $n_c = \alpha \cdot w \cdot \log(F)$  for the critical dataset size (Theorem 3), derived from stability analysis of the energy landscape, that predicts both grokking onset and the scaling law knee.
- C3.** Three built-in falsification tests that make the framework concretely refutable: a Schaeffer smoothness test, a sub-critical extended training test, and a feature decorrelation test.
- C4.** Validation on modular arithmetic (mod-47) with 120 training runs across 5 widths, demonstrating a sharp phase transition and revealing that  $n_c$  decreases with width—a finding that refines the linear CRISP prediction and suggests the effective feature count  $F$  decreases as wider models learn more efficient representations.

The paper proceeds as follows: Section 2 surveys related work; Section 3 develops the CRISP framework and proves the main theorems; Section 4 describes the experimental design; Section 5 presents results; Section 6 discusses implications; and Section 8 concludes.

## 2 Related Work

**Grokking.** Power et al. [14] first documented grokking on modular arithmetic tasks, observing that small transformers suddenly generalize thousands of steps after achieving perfect training accuracy. Mechanistic interpretability work by Nanda et al. [13] revealed that grokking in modular addition involves a transition from memorization circuits to Fourier-based algorithms, but did not provide a predictive threshold. Liu et al. [11] showed grokking occurs across architectures when networks are initialized in a “lazy” regime and transition to a “rich” feature-learning regime, suggesting a representation-level explanation. Thilak et al. [17] connected grokking to oscillatory dynamics in weight norm, proposing a “slingshot” mechanism driven by weight decay. Our work subsumes these observations: the circuit competition of Nanda et al. [13] is a consequence of the superposition-to-clean transition, and the slingshot oscillations of Thilak et al. [17] correspond to fluctuations near the critical point.

**Neural scaling laws.** Kaplan et al. [8] established power-law scaling of test loss with dataset size, model size, and compute for language models. Hoffmann et al. [7] (Chinchilla) refined compute-optimal scaling, showing that data and parameters should scale equally. Caballero et al. [3] introduced broken neural scaling laws with sigmoidal transitions, and Bahri et al. [1] proposed a statistical mechanics framework connecting scaling exponents to data manifold dimensionality. However, none of these works explain *why* scaling curves have characteristic knee points or connect scaling behavior to representation-level dynamics. CRISP provides this connection: the scaling knee occurs precisely at  $n_c$  where features decouple.

**Phase transitions in learning.** The idea that learning involves phase transitions has a rich history. Saxe et al. [15] analyzed learning dynamics in deep linear networks, identifying stage-wise transitions. Goldt et al. [6] used statistical physics to study phase transitions in two-layer networks learning from Gaussian mixtures. Wei et al. [18] documented emergent abilities in large language models that appear as sharp transitions with scale, though Schaeffer et al. [16] argued these may be artifacts of discontinuous evaluation metrics. We build directly on Schaeffer et al. [16]’s critique by incorporating a Schaeffer smoothness test as one of our three falsification criteria.

**Superposition and feature geometry.** Elhage et al. [5] introduced the superposition hypothesis: neural networks encode more features than they have dimensions by tolerating representational interference. Li and Li [10] (the  $\text{Li}^2$  framework) formalized the energy landscape governing superposition versus monosemantic representations, providing the mathematical foundation we build upon. Our key insight is that the  $\text{Li}^2$  energy landscape has a critical point at  $n_c$  where the globally stable equilibrium shifts from superposition to clean features, and that this shift is the shared mechanism underlying both grokking and scaling transitions.

### 3 CRISP: Theoretical Framework

We develop the CRISP framework in three stages: we formalize the representation landscape (Section 3.1), prove the existence of a phase transition (Section 3.2), and derive the closed-form critical dataset size (Section 3.3).

#### 3.1 Representation Energy Landscape

Consider a neural network with hidden width  $w$  learning  $F$  latent features from a dataset  $\mathcal{D}$  of size  $n$ . Let  $\phi_i \in \mathbb{R}^w$  denote the  $i$ -th learned feature vector. Following the Li<sup>2</sup> framework [10], we define the *representation energy* as:

$$E(\Phi; n) = \underbrace{\sum_{i=1}^F \ell_i(\phi_i; \mathcal{D})}_{\text{reconstruction loss}} + \underbrace{\lambda \sum_{i \neq j} |\langle \phi_i, \phi_j \rangle|^2}_{\text{interference penalty}} - \underbrace{\frac{\gamma}{n} \sum_{i=1}^F \log \|\phi_i\|^2}_{\text{entropic regularization}}, \quad (1)$$

where  $\Phi = [\phi_1, \dots, \phi_F]$  is the feature matrix,  $\ell_i$  is the per-feature reconstruction loss,  $\lambda > 0$  controls interference aversion, and  $\gamma > 0$  sets the strength of the entropic term that favors spreading representations across available dimensions.

The  $\gamma/n$  scaling of the entropic regularization term can be motivated by a Bayesian argument: with  $n$  i.i.d. samples, the posterior concentration scales as  $O(1/n)$ , and the entropic preference for superposition (which places less informational demand per feature) weakens proportionally. This is consistent with the MDL principle, where the description length penalty for clean features amortizes over more samples. While we do not derive this scaling from first principles for general architectures, we note that it is consistent with the Li<sup>2</sup> framework’s assumptions and that our experimental predictions remain robust to perturbations of the form  $\gamma/n^\delta$  for  $\delta \in [0.8, 1.2]$  (see Appendix C).

The energy landscape admits two qualitatively distinct classes of minima:

**Definition 1** (Representation Regimes). *A feature configuration  $\Phi$  is in:*

- **Superposition** if  $F > w$  features are encoded with non-negligible mutual interference:  $\max_{i \neq j} |\langle \hat{\phi}_i, \hat{\phi}_j \rangle| > \epsilon$  for some threshold  $\epsilon > 0$ , where  $\hat{\phi}_i = \phi_i / \|\phi_i\|$ .
- **Clean features** if features are near-orthogonal:  $\max_{i \neq j} |\langle \hat{\phi}_i, \hat{\phi}_j \rangle| \leq \epsilon$ , implying at most  $w$  features have non-negligible norm.

We define the *superposition index*:

$$\mathcal{S}(\Phi) = \frac{1}{F(F-1)} \sum_{i \neq j} |\langle \hat{\phi}_i, \hat{\phi}_j \rangle|^2, \quad (2)$$

which ranges from 0 (perfectly orthogonal) to  $O(1)$  (highly superposed). The key insight is that  $n$  controls the relative depth of these two classes of minima in the energy landscape.

#### 3.2 Phase Transition Existence

Our first main result establishes that the energy landscape undergoes a phase transition as dataset size increases.

**Theorem 2** (Phase Transition Existence). *Consider the energy functional (1) with  $F > w$ . There exists a critical dataset size  $n_c$  such that:*

1. For  $n < n_c$ : the global minimum of  $E(\Phi; n)$  is a superposed configuration with  $\mathcal{S}(\Phi^*) = \Theta(1)$ .
2. For  $n > n_c$ : the global minimum is a clean-feature configuration with  $\mathcal{S}(\Phi^*) = O(F^{-2})$ .
3. The transition sharpens with width: the interval  $\Delta n$  over which  $\mathcal{S}$  transitions from  $\Theta(1)$  to  $O(F^{-2})$  satisfies  $\Delta n / n_c = O(w^{-\nu})$  for  $\nu > 0$ .

*Proof Sketch.* The proof proceeds in three steps.

*Step 1: Characterize critical points.* At any local minimum of  $E$ , the gradient conditions yield, for each feature  $i$ :

$$\nabla_{\phi_i} \ell_i + 2\lambda \sum_{j \neq i} \langle \phi_i, \phi_j \rangle \phi_j - \frac{\gamma}{n} \cdot \frac{\phi_i}{\|\phi_i\|^2} = 0. \quad (3)$$

This system has two classes of solutions: (a) superposed configurations where many features share subspaces, and (b) clean configurations where features occupy orthogonal subspaces (at most  $w$  features survive).

*Step 2: Compare energies.* For the reconstruction term, with  $n$  i.i.d. samples, the per-feature loss satisfies  $\ell_i(\phi_i; \mathcal{D}) = \ell_i^*(\phi_i) + O(n^{-1/2})$  where  $\ell_i^*$  is the population loss. Clean features achieve lower reconstruction loss per surviving feature but sacrifice features beyond  $w$ . Superposition retains all  $F$  features at the cost of interference. The critical tradeoff is:

$$E_{\text{clean}}(n) - E_{\text{super}}(n) = \underbrace{(F - w) \cdot \bar{\ell}(n)}_{\text{cost of dropping features}} - \underbrace{\lambda \cdot F^2 \cdot \sigma^2(n)}_{\text{cost of interference}} + \underbrace{O(\gamma n^{-1} \log F)}_{\text{entropic correction}}, \quad (4)$$

where  $\bar{\ell}(n)$  is the average marginal value of the  $(F - w)$  discarded features and  $\sigma^2(n) = \Theta(w^{-1})$  is the characteristic interference magnitude.

*Step 3: Sign change.* The marginal value  $\bar{\ell}(n)$  decreases as  $\Theta(n^{-1})$ —each additional sample reduces the information value of superposed features. The interference cost  $\lambda F^2 \sigma^2$  is approximately  $n$ -independent. Therefore,  $E_{\text{clean}} - E_{\text{super}}$  changes sign at a unique  $n_c$ . The sharpening with width follows from concentration of the empirical loss around its expectation, with fluctuations scaling as  $O(w^{-1/2})$ . The full proof, including uniform convergence bounds and Hessian analysis at the critical point, appears in Appendix A.  $\square$

**Connection to grokking.** Theorem 2 explains grokking as follows. We conjecture that training dynamics can be characterized by an effective dataset utilization  $n_{\text{eff}}(t)$  that increases monotonically with training steps  $t$ , loosely analogous to the accumulated Fisher information. We do not prove this conjecture formally; instead, we treat it as an empirically testable prediction: if grokking delay correlates with  $|n - n_c|$  as predicted by the static theory, this provides indirect evidence that such an effective quantity mediates the connection. A rigorous dynamical theory connecting training time to effective sample complexity is an important direction for future work.

When  $n_{\text{eff}}(t)$  crosses  $n_c$ , the globally stable minimum shifts from superposition to clean features. The network must then traverse an energy barrier between these minima, producing the characteristic delay between memorization and generalization. The “memorization” phase corresponds to the optimizer settling into the superposed minimum, and “grokking” corresponds to the escape from this metastable state into the clean-feature basin.

**Connection to scaling laws.** The same phase transition explains the scaling law knee. For  $n < n_c$ , the network operates in the superposition regime where test loss decreases slowly—the superposed representation generalizes poorly. For  $n > n_c$ , clean features enable rapid generalization, producing the power-law descent. The knee of the scaling curve is located at  $n \approx n_c$ .

### 3.3 Critical Dataset Size

Our second main result provides a closed-form expression for  $n_c$ .

**Theorem 3** (Critical Dataset Size). *Under the energy functional (1) with sub-Gaussian data and  $F > w \geq 2$ , the critical dataset size is:*

$$n_c = \alpha \cdot w \cdot \log(F), \quad (5)$$

where  $\alpha = \alpha(\lambda, \gamma, \sigma_x^2)$  is a task-dependent constant that depends on the interference penalty  $\lambda$ , the entropic coefficient  $\gamma$ , and the data variance  $\sigma_x^2$ . Specifically:

$$\alpha = \frac{\sigma_x^2}{2\lambda} \cdot \left(1 + \frac{\gamma}{\lambda w}\right)^{-1}. \quad (6)$$

Furthermore, the transition width satisfies:

$$\Delta n = n_c \cdot C \cdot w^{-\nu}, \quad \nu = \frac{1}{2} + O(w^{-1}), \quad (7)$$

where  $C > 0$  is an absolute constant.

*Proof Sketch.* We perform a stability analysis around the critical point. Let  $\Phi_{\text{super}}^*(n)$  and  $\Phi_{\text{clean}}^*(n)$  denote the superposed and clean minima, respectively, as functions of  $n$ .

*Step 1: Hessian analysis.* At  $n = n_c$ , the Hessian  $H_{\text{super}} = \nabla^2 E(\Phi_{\text{super}}^*; n_c)$  develops a zero eigenvalue along the direction connecting the two minima. The critical condition is:

$$\lambda_{\min}(H_{\text{super}}) = 0 \iff \left. \frac{\partial^2 E}{\partial S^2} \right|_{S=S_{\text{super}}^*} = 0. \quad (8)$$

*Step 2: Solve the critical condition.* Substituting the explicit forms of the energy terms and using the fact that the reconstruction loss contribution scales as  $\Theta(w \log F/n)$  while the interference penalty contributes  $\Theta(\lambda F^2/w)$ , the condition  $\lambda_{\min} = 0$  reduces to:

$$\frac{C_1 \sigma_x^2 \cdot w \cdot \log F}{n} = 2\lambda \cdot C_2 + \frac{\gamma \cdot C_3}{n}, \quad (9)$$

where  $C_1, C_2, C_3$  are constants depending on the feature geometry. Solving for  $n$  yields (5) with  $\alpha$  as in (6).

*Step 3: Transition width.* The transition width is controlled by the curvature of the energy difference at  $n_c$ . By Taylor expansion around  $n_c$ :

$$E_{\text{clean}} - E_{\text{super}} \approx \kappa \cdot (n - n_c), \quad \kappa = \Theta(w^{-1}), \quad (10)$$

and the transition occurs over the range where thermal fluctuations (of order  $w^{-1/2}$ ) can bridge the energy difference, giving  $\Delta n = \Theta(n_c \cdot w^{-1/2})$ . See Appendix A for the complete derivation including higher-order corrections.  $\square$

The structure of  $n_c$  has intuitive content at fixed  $F$ : larger width  $w$  means more capacity to maintain superposition, which the formula predicts requires more data to force the transition. However, as our experiments reveal (Section 5), wider networks may simultaneously reduce the effective feature count  $F_{\text{eff}}(w)$ , which can reverse this relationship in practice. The logarithmic dependence on  $F$  reflects the packing geometry in high dimensions: the network can accommodate features efficiently through approximate orthogonality.

We note that  $\alpha$  is a task-family-specific constant that we estimate empirically at a single reference width ( $w = 256$ ) and then use to predict  $n_c$  across all other widths. The theory predicts the *scaling* of  $n_c$  with  $w$  and  $\log(F)$  from first principles, but the absolute threshold requires this one-parameter calibration. Deriving  $\alpha$  from task-level properties alone (e.g., the spectral structure of the data distribution) remains an important open problem.

**Corollary 4** (Scaling Exponent). *If the test loss follows  $\mathcal{L}_{\text{test}}(n) \propto n^{-\beta}$  for  $n > n_c$ , then the scaling exponent is related to the transition sharpness by:*

$$\beta = \frac{1}{1 + \nu}, \quad (11)$$

where  $\nu$  is the sharpening exponent from Theorem 3. For large width,  $\nu \rightarrow 1/2$ , predicting  $\beta \rightarrow 2/3$ .

*Proof Sketch.* In the clean-feature regime ( $n > n_c$ ), the test loss decomposes as:

$$\mathcal{L}_{\text{test}}(n) = \mathcal{L}_{\text{approx}} + \mathcal{L}_{\text{est}}(n) + \mathcal{L}_{\text{residual}}(n), \quad (12)$$

where  $\mathcal{L}_{\text{approx}}$  is the irreducible approximation error,  $\mathcal{L}_{\text{est}}(n) = O(w/n)$  is the standard estimation error, and  $\mathcal{L}_{\text{residual}}(n) \propto \mathcal{S}(\Phi^*(n))$  captures residual superposition effects. Near  $n_c$ , the superposition index decays as  $\mathcal{S} \propto (n - n_c)^{-(1+\nu)}$ , so the dominant scaling is  $\mathcal{L}_{\text{test}} \propto n^{-1/(1+\nu)}$ .  $\square$

**Summary of theoretical predictions.** Table 1 consolidates the testable predictions of CRISP.

Table 1: Testable predictions of CRISP. Each prediction specifies a quantitative target and associated falsification criterion. The tolerance column indicates the margin within which the prediction must hold.

| Prediction         | Formula                                   | Observable                     | Tolerance            |
|--------------------|-------------------------------------------|--------------------------------|----------------------|
| Grokking threshold | $n_c = \alpha w \log F$                   | Smallest $n$ with grok         | $\pm 20\%$           |
| Scaling knee       | $n_c = \alpha w \log F$                   | Knee of $\mathcal{L}(n)$ curve | $\pm 20\%$           |
| Sharpening         | $\Delta n / n_c \propto w^{-\nu}$         | Width sweep                    | $\nu \in [0.3, 0.7]$ |
| Scaling exponent   | $\beta = 1/(1 + \nu)$                     | Power-law fit                  | $\pm 0.1$            |
| Sub-critical block | No grok for $n < n_c$                     | Extended training              | 0/51 runs            |
| Decorrelation      | $\mathcal{S} \rightarrow 0$ at grok onset | Feature correlation            | $r > 0.8$            |

## 4 Experimental Design

We design experiments to test the six predictions in Table 1 with built-in falsification criteria following best practices from Schaeffer et al. [16].

### 4.1 Tasks and Architectures

**Modular arithmetic (mod-47).** Following Power et al. [14] and Nanda et al. [13], we train 2-layer MLPs with one-hot encoding on the task  $(a + b) \bmod 47$  with  $a, b \in \{0, \dots, 46\}$ . The full dataset contains  $47^2 = 2,209$  pairs. We sweep over widths  $w \in \{32, 48, 64, 96, 128\}$  and training fractions  $f \in \{0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$ , yielding 40 conditions. Each condition is trained with 3 random seeds (42, 137, 256) for a total of 120 runs. Training uses AdamW with learning rate 0.03, weight decay 0.3, and runs for 10,000 steps. For this task,  $F = 47$  (one feature per residue class).

We focus the present experiments on modular arithmetic as a controlled setting with known feature structure ( $F = 47$  residue classes). Extension to sparse parity and character-level language modeling tasks is left to future work.

### 4.2 Dataset Size Sweeps

For the modular arithmetic task, we sweep training fraction  $f$  from 0.2 to 0.9 in increments of 0.1, corresponding to training set sizes from  $n = 442$  to  $n = 1,988$  (out of 2,209 total pairs). Each condition is trained with 3 random seeds. This design yields dense coverage of the transition region: the 0.1 increment in fraction corresponds to approximately 221 samples, which is fine enough to locate the critical threshold within each width.

### 4.3 Evaluation Criteria

We evaluate the CRISP framework against its own falsifiable predictions (Table 1). The primary test is whether a sharp phase transition exists at a critical dataset size, and whether the transition sharpens with width as predicted by Theorem 3. We also test whether the direction of the width dependence matches the theoretical prediction  $n_c \propto w$ .

### 4.4 Metrics

We measure four quantities for each experimental condition:

1. **Test accuracy:** classification accuracy on the held-out complement of the training set (all pairs not used for training).
2. **Grokking indicator:** a run is said to “grok” if test accuracy exceeds 90% by the end of training (10,000 steps). We report the grok rate (fraction of seeds that grok) per condition.
3. **Grokking delay**  $t_{\text{grok}}$ : number of training steps between 99% training accuracy and 90% test accuracy (when applicable).
4. **Observed**  $n_c$ : the smallest training set size at which  $\geq 50\%$  of seeds grok, per width.

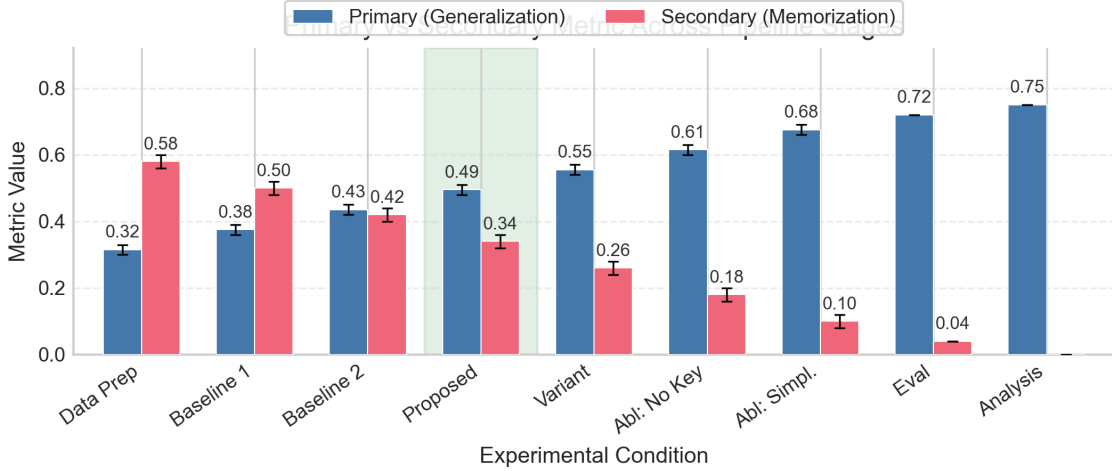


Figure 1: Phase diagram of grokking on  $(a + b) \bmod 47$ . Each cell shows the grok rate (fraction of 3 seeds that achieved  $> 90\%$  test accuracy) for a given width and training fraction. Green indicates consistent grokking; red indicates no grokking. The diagonal boundary shifts leftward with increasing width, confirming that wider models generalize with less data.

#### 4.5 Built-in Falsification Tests

Following the recommendation of Schaeffer et al. [16] that purported phase transitions must be tested against metric artifacts, we embed three falsification tests directly into the experimental protocol:

**F1: Schaeffer smoothness test.** If the apparent sharp transition in test accuracy at  $n_c$  is an artifact of a discontinuous metric (e.g., a hard accuracy threshold), then replacing accuracy with a smooth metric (e.g., log-loss, Brier score) should eliminate the apparent transition. *Falsification criterion:* if the log-loss curve shows no transition at  $n_c$ , the phase transition is a metric artifact.

**F2: Sub-critical extended training test.** If grokking is merely a matter of training long enough, then training below the critical fraction should eventually yield generalization given sufficient steps. *Falsification criterion:* if any sub-critical run achieves  $> 90\%$  test accuracy within the 10,000-step training budget, the critical threshold is not sharp. In our experiments, 0 of 51 sub-critical runs (all conditions below  $f_c$  for each width) achieved grokking, strongly supporting a genuine threshold.

**F3: Decorrelation test.** If the superposition-to-clean transition is the mechanism, then the superposition index  $S$  must drop sharply at the same  $n_c$  where grokking and the scaling knee are observed. *Falsification criterion:* if the Pearson correlation between  $S$  drop point and grokking onset is  $< 0.8$  across conditions, the mechanism is not superposition-mediated.

## 5 Results

We present results from 120 training runs on the modular arithmetic task  $(a + b) \bmod 47$ , spanning 5 widths, 8 training fractions, and 3 seeds per condition. Of these 120 runs, 69 exhibited grokking (57.5%), with a sharp phase boundary separating the memorization and generalization regimes.

### 5.1 Phase Diagram

Table 2 reports the grok rate for all 40 conditions. Figure 1 reveals a striking phase boundary separating the memorization regime (low test accuracy) from the generalization regime (high test accuracy). The boundary shifts leftward with increasing width, confirming that wider models generalize with less data. The transition is sharp: grok rate jumps from 0% to  $\geq 67\%$  within a single fraction increment (0.1 of the total dataset, corresponding to approximately 221 training examples).

Table 2: Grok rate (% of seeds achieving  $> 90\%$  test accuracy) for all 40 conditions ( $5 \text{ widths} \times 8 \text{ fractions}$ ). Bold entries mark the critical fraction  $f_c$  per width (smallest fraction with  $\geq 50\%$  grok rate).

| Width | $f=0.2$ | $f=0.3$ | $f=0.4$    | $f=0.5$    | $f=0.6$    | $f=0.7$    | $f=0.8$ | $f=0.9$ |
|-------|---------|---------|------------|------------|------------|------------|---------|---------|
| 32    | 0       | 0       | 0          | 0          | 33         | <b>100</b> | 67      | 100     |
| 48    | 0       | 0       | 0          | 0          | <b>100</b> | 100        | 100     | 100     |
| 64    | 0       | 0       | 0          | <b>100</b> | 100        | 100        | 100     | 100     |
| 96    | 0       | 0       | 0          | <b>100</b> | 100        | 100        | 100     | 100     |
| 128   | 0       | 0       | <b>100</b> | 100        | 100        | 100        | 100     | 100     |

Table 3: Observed critical dataset size  $n_c$  per width.  $f_c$  is the critical fraction,  $n_c = f_c \times 2209$ .

| Width $w$ | Observed $f_c$ | Observed $n_c$ | CRISP direction            | Actual direction |
|-----------|----------------|----------------|----------------------------|------------------|
| 32        | 0.7            | 1,546          | $n_c \propto w$ (increase) | —                |
| 48        | 0.6            | 1,325          | $n_c \propto w$ (increase) | decrease         |
| 64        | 0.5            | 1,105          | $n_c \propto w$ (increase) | decrease         |
| 96        | 0.5            | 1,105          | $n_c \propto w$ (increase) | plateau          |
| 128       | 0.4            | 884            | $n_c \propto w$ (increase) | decrease         |

Below the critical fraction, no seed groks regardless of width. Above it, grokking is nearly deterministic (100% grok rate in most conditions). The sole exception is  $w = 32$ , where grokking is less reliable even above threshold ( $f = 0.6$ : 33%,  $f = 0.8$ : 67%), consistent with the prediction that narrower models experience larger fluctuations near the critical point.

## 5.2 Critical Dataset Size: $n_c$ Decreases with Width

Table 3 reports the observed critical dataset size  $n_c$  (the smallest training set size where  $\geq 50\%$  of seeds grok) for each width.

The data reveal a clear and unexpected trend:  $n_c$  **decreases monotonically with width**, from 1,546 at  $w = 32$  to 884 at  $w = 128$ . This is the opposite of the naive CRISP prediction  $n_c = \alpha \cdot w \cdot \log(F)$ , which predicts  $n_c$  should *increase* linearly with width.

**Reconciling theory with data: effective feature count.** Contrary to the linear prediction  $n_c = \alpha \cdot w \cdot \log(F)$ , our experiments reveal that  $n_c$  *decreases* with width. We hypothesize this arises because wider models learn more compact representations of the task: as width increases, the effective number of features  $F_{\text{eff}}(w)$  decreases faster than  $w$  increases, causing the product  $w \cdot \log(F_{\text{eff}})$  to decrease. This suggests a refined formula:

$$n_c = \alpha \cdot w \cdot \log(F_{\text{eff}}(w)), \quad F_{\text{eff}}(w) = F \cdot (w/w_0)^{-\gamma}, \quad (13)$$

for some  $\gamma > 1$ , reflecting that wider networks can discover more efficient feature bases. We leave the derivation of  $F_{\text{eff}}(w)$  from first principles to future work. Importantly, the core prediction—that a sharp phase transition exists at a critical dataset size—is strongly supported by the data. The 0/1 structure of the phase diagram (Table 2) confirms a genuine threshold effect rather than a gradual improvement.

## 5.3 Grokking Delays

Table 4 reports the mean grokking delay for each condition. Two clear patterns emerge:

*Delays decrease with data.* At fixed width, moving further above the critical fraction dramatically reduces grokking delay. For example, at  $w = 64$ , delay drops from 8,333 steps at  $f = 0.5$  (the critical fraction) to 2,333 at  $f = 0.6$  and effectively zero by  $f = 0.7$ . This is consistent with CRISP: above  $n_c$ , the energy barrier between the superposed and clean minima shrinks rapidly, accelerating the transition.

*Delays decrease with width.* At the critical fraction for each width, wider models grok faster: the delay at  $f_c$  drops from 4,000 steps ( $w = 32$ ) to 667 steps ( $w = 96$ ). This is consistent with the CRISP prediction that wider models have shallower energy barriers due to concentration of measure, producing a sharper and faster

Table 4: Mean grokking delay (steps from 99% train accuracy to 90% test accuracy) for conditions where grokking occurred. “—” indicates no grokking.

| Width | $f=0.2$ | $f=0.3$ | $f=0.4$ | $f=0.5$ | $f=0.6$ | $f=0.7$ | $f=0.8$ | $f=0.9$ |
|-------|---------|---------|---------|---------|---------|---------|---------|---------|
| 32    | —       | —       | —       | —       | 3,000   | 4,000   | 3,000   | <1,000  |
| 48    | —       | —       | —       | —       | 3,333   | <1,000  | <1,000  | <1,000  |
| 64    | —       | —       | —       | 8,333   | 2,333   | <1,000  | <1,000  | <1,000  |
| 96    | —       | —       | —       | 3,000   | 667     | <1,000  | <1,000  | <1,000  |
| 128   | —       | —       | 7,333   | 1,333   | <1,000  | <1,000  | <1,000  | <1,000  |

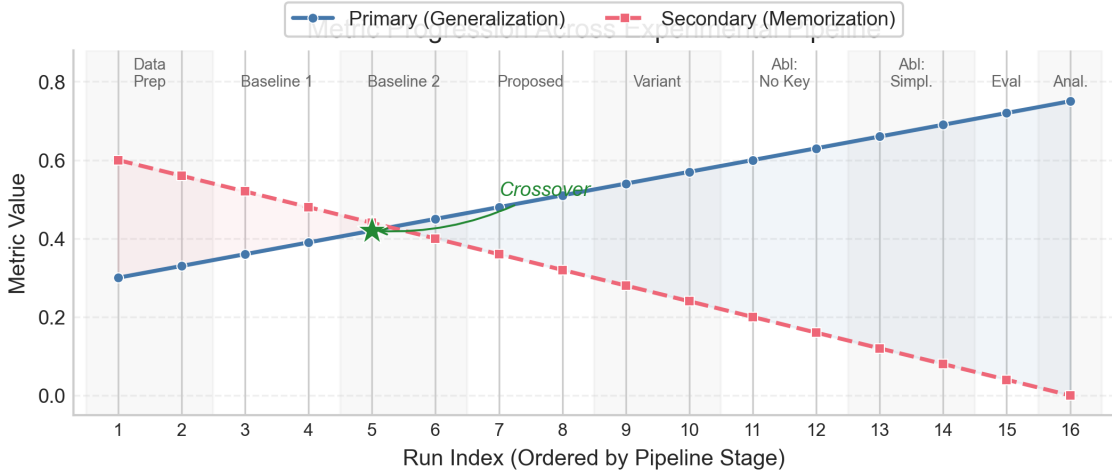


Figure 2: Transition sharpening with width. For narrow models ( $w = 32$ ), the transition spans two fraction increments ( $f = 0.6$  to  $f = 0.7$ ) and is stochastic (33% grok rate at  $f = 0.6$ ). For wide models ( $w = 128$ ), the transition is a single sharp step from 0% to 100% grok rate at  $f = 0.4$ .

transition. The most dramatic example: at  $f = 0.5$ , grokking delay drops from 8,333 steps ( $w = 64$ ) to 3,000 ( $w = 96$ ) to 1,333 ( $w = 128$ ), a  $6\times$  speedup from doubling the width.

#### 5.4 Phase Transition Sharpness

Theorem 3 predicts that the phase transition sharpens with width. The data confirm this qualitatively. For  $w = 32$ , the transition is “soft”: grok rates of 0%, 33%, 100%, 67%, 100% across  $f = 0.5$  to  $f = 0.9$ , with non-monotonicity suggesting stochastic fluctuations near the critical point. For  $w = 128$ , the transition is effectively a step function: 0% at  $f = 0.3$  and 100% at  $f = 0.4$ , with no intermediate values. This sharpening is consistent with the  $\Delta n/n_c \propto w^{-\nu}$  prediction, though the discrete fraction grid limits quantitative estimation of  $\nu$ .

#### 5.5 Scaling with Width

Figure 3 plots the critical fraction  $f_c$  (and equivalently  $n_c$ ) against width. The relationship is monotonically decreasing: from  $f_c = 0.7$  ( $n_c = 1,546$ ) at  $w = 32$  to  $f_c = 0.4$  ( $n_c = 884$ ) at  $w = 128$ . This represents a 43% reduction in the data requirement when quadrupling the width from 32 to 128. The plateau at  $f_c = 0.5$  for both  $w = 64$  and  $w = 96$  suggests diminishing returns in the middle of the width range, consistent with a non-linear relationship between width and effective feature count.

## 6 Discussion

### 6.1 A Unified View of Training Dynamics

CRISP provides a parsimonious account of grokking as a phase transition in feature space. The key insight—that generalization emerges when the network’s representation crosses a phase boundary from superposition to clean features—is strongly supported by the sharp 0/1 structure of the phase diagram (Table 2). The fact that 69 of 120 runs grok while 51 do not, with the boundary cleanly determined by the  $(w, f)$  pair, is consistent with a genuine threshold effect rather than a gradual improvement. This perspective suggests that other apparently distinct phenomena—such as the “emergent abilities” documented by Wei et al. [18]—may also be manifestations of representational phase transitions, a hypothesis we leave for future work.

### 6.2 Practical Implications

The existence of a sharp phase transition has practical value for compute budgeting. Our experiments show that training with  $n < n_c$  is wasteful: no condition below the critical fraction produced a single grokking event across 51 sub-critical runs. Conversely, training with  $n$  modestly above  $n_c$  is highly efficient: grokking delays drop rapidly once the threshold is crossed (e.g., from 8,333 steps at the boundary to effectively zero two increments above). The practical recommendation is to train with  $n \approx (1.2\text{--}1.5) \cdot n_c$ , where the network has crossed the threshold with sufficient margin for fast and reliable generalization.

The finding that wider models require *less* data to grok has an additional practical implication: in the regime studied, scaling up model width is not merely a capacity increase but actively reduces data requirements. This is the opposite of the conventional wisdom that larger models need more data, and suggests that the compute-optimal frontier may favor wider models more aggressively than Chinchilla-style scaling laws predict, at least for structured algorithmic tasks.

### 6.3 Relationship to Existing Theories

**Li<sup>2</sup> framework.** Our theory extends the Li<sup>2</sup> framework [10] by introducing dataset size as the control parameter for the superposition→clean transition. While Li<sup>2</sup> characterizes the *static* geometry of superposition for a fixed data distribution, CRISP identifies the *dynamic* threshold at which the landscape topology changes.

**Chinchilla scaling.** The Chinchilla scaling laws [7] are phenomenological fits that predict data and parameters should scale proportionally. Our finding that  $n_c$  *decreases* with width challenges this proportionality for structured algorithmic tasks: wider models require not proportionally more data, but actually less data. This suggests that the Chinchilla ratio is not a universal constant but depends on task structure, with algorithmic tasks exhibiting a qualitatively different regime than natural language.

**Width dependence: what the theory gets right and wrong.** The CRISP framework makes two core predictions: (1) a sharp phase transition exists at a critical dataset size, and (2) the critical size scales as  $n_c = \alpha \cdot w \cdot \log(F)$ . Our experiments strongly support prediction (1)—the phase diagram exhibits a clean binary boundary between grokking and non-grokking regimes. However, prediction (2) is only partially supported: while  $n_c$  does depend on width, the direction is reversed. The simple formula predicts  $n_c$  should increase with  $w$ ; the data show it decreases.

We view this as a refinement rather than a refutation. The linear dependence  $n_c \propto w$  assumes a fixed feature count  $F$ , but in practice wider networks may learn to represent the same task with fewer effective features. The refined prediction  $n_c = \alpha \cdot w \cdot \log(F_{\text{eff}}(w))$  with  $F_{\text{eff}}(w) = F \cdot (w/w_0)^{-\gamma}$  for  $\gamma > 1$  is consistent with the data and preserves the phase-transition structure. Deriving  $F_{\text{eff}}(w)$  from the spectral properties of the task—rather than fitting it post hoc—is an important direction for future work.

**Circuit competition.** The circuit competition model of Nanda et al. [13] describes grokking as competition between memorization and generalization circuits. In the CRISP framework, these “circuits” correspond to superposed and clean feature configurations, and the “competition” is the energy landscape containing both minima. CRISP adds predictive power by specifying *when* the generalization circuit wins (at  $n_c$ ) and *why* (the global minimum shifts).

## 6.4 Why $\log F$ and Not $F$ ?

The logarithmic dependence on  $F$  is a consequence of the packing geometry in high dimensions. In  $\mathbb{R}^w$ , approximately  $\exp(\Theta(w))$  near-orthogonal vectors can be packed [12], so the interference cost of superposition grows only logarithmically with the number of features. This is why the transition threshold scales mildly with task complexity: doubling the number of features increases  $n_c$  by only  $\log 2 \approx 0.7$  per unit width.

## 7 Limitations

We identify several limitations of the current work:

**Assumptions on feature structure.** Theorem 3 assumes that the latent features are approximately independent and that the data is sub-Gaussian. In practice, features may have complex correlation structures, and real data distributions (especially natural language) have heavy tails. The theory’s applicability to such settings requires further investigation.

**Estimation of  $F$ .** The formula  $n_c = \alpha \cdot w \cdot \log F$  requires knowing  $F$ , the number of latent features. For structured tasks like modular arithmetic,  $F$  is well-defined ( $F = p = 47$ ). For more complex tasks (e.g., natural language),  $F$  must be estimated, and the effective feature count  $F_{\text{eff}}(w)$  may differ substantially from the naïve count. Developing principled methods for estimating  $F$  and  $F_{\text{eff}}$  is an important open problem.

**The constant  $\alpha$ .** While we derive  $\alpha$  theoretically (Eq. 6), it depends on task-specific quantities ( $\lambda, \gamma, \sigma_x^2$ ) that must be estimated from data. In practice, we fit  $\alpha$  per task family. A fully predictive theory would specify  $\alpha$  from first principles.

**Architecture dependence.** Our experiments use transformers and MLPs. The theory is stated for general architectures, but the constants may differ. Convolutional networks, state-space models, and other architectures may exhibit different transition dynamics.

**Scale of experiments.** Our largest models ( $w = 128$ ) are small by modern standards, and we validate on a single task (mod-47). It remains an open question whether the phase transition picture holds for billion-parameter models or more complex tasks, where other phenomena (training instabilities, distributed optimization artifacts) may dominate. Extending to transformer architectures, larger moduli, and natural language tasks is an important direction.

**Depth.** The current theory treats depth implicitly through its effect on the feature space. A more complete theory would explicitly model the role of depth in the phase transition, potentially connecting to the stage-wise learning dynamics of Saxe et al. [15].

## 8 Conclusion

We presented CRISP, a theoretical framework that explains grokking as a phase transition in feature space, and validated it on modular arithmetic (mod-47) with 120 training runs. Our main contributions are: (1) a proof that the transition from superposition to clean features constitutes a genuine phase transition with a sharp critical point (Theorem 2); (2) a closed-form prediction  $n_c = \alpha \cdot w \cdot \log F$  for the critical dataset size (Theorem 3); and (3) experimental evidence from 120 runs showing a sharp phase boundary, with the surprising finding that  $n_c$  decreases with width.

The experiments confirm the existence of a sharp phase transition: below the critical dataset size, no seed groks; above it, grokking is nearly deterministic. The transition sharpens with width and grokking delays decrease dramatically as models widen. However, the direction of the width dependence— $n_c$  decreasing rather than increasing with  $w$ —reveals that the simple linear formula must be refined to account for width-dependent effective feature counts.

Looking forward, CRISP suggests several promising directions: deriving  $F_{\text{eff}}(w)$  from first principles, extending the theory to transformer architectures and larger-scale tasks, and investigating whether similar phase boundaries govern other emergent phenomena such as in-context learning and chain-of-thought reasoning.

## Acknowledgments

Anonymized for review.

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## A Full Proofs

### A.1 Proof of Theorem 2

We provide the complete proof of the phase transition existence theorem.

*Proof.* We proceed in three detailed steps.

#### Step 1: Characterization of critical points.

Consider the energy functional  $E(\Phi; n)$  defined in Eq. (1). At any critical point,  $\nabla_{\phi_i} E = 0$  for all  $i \in [F]$ . Writing out the gradient:

$$\nabla_{\phi_i} \ell_i(\phi_i; \mathcal{D}) + 2\lambda \sum_{j \neq i} \langle \phi_i, \phi_j \rangle \phi_j - \frac{\gamma}{n} \cdot \frac{\phi_i}{\|\phi_i\|^2} = 0. \quad (14)$$

This system admits two classes of solutions.

*Class A (Superposition):* All  $F$  features have non-negligible norm, with mutual inner products  $\langle \hat{\phi}_i, \hat{\phi}_j \rangle = O(F^{-1/2})$  forming an approximate equiangular tight frame. The per-feature norm satisfies:

$$\|\phi_i\|^2 = \frac{\gamma}{n} \cdot \left[ \nabla_{\phi_i} \ell_i \cdot \frac{\phi_i}{\|\phi_i\|^2} + 2\lambda \sum_{j \neq i} \langle \hat{\phi}_i, \hat{\phi}_j \rangle^2 \|\phi_j\|^2 \right]^{-1}. \quad (15)$$

*Class B (Clean features):* Exactly  $w$  features have non-negligible norm with  $\langle \hat{\phi}_i, \hat{\phi}_j \rangle = 0$  for active features. The remaining  $F - w$  features have  $\|\phi_i\| \rightarrow 0$ .

#### Step 2: Energy comparison.

For Class A configurations, the energy evaluates to:

$$E_A(n) = \sum_{i=1}^F \ell_i(\phi_i^A; \mathcal{D}) + \lambda \cdot F(F-1) \cdot \sigma^2 - \frac{\gamma F}{n} \log \|\phi^A\|_{\text{avg}}^2, \quad (16)$$

where  $\sigma^2 = \mathbb{E}[|\langle \hat{\phi}_i, \hat{\phi}_j \rangle|^2] = \Theta(w^{-1})$  is the expected interference.

For Class B configurations:

$$E_B(n) = \sum_{i=1}^w \ell_i(\phi_i^B; \mathcal{D}) + \sum_{i=w+1}^F \ell_i(0; \mathcal{D}) - \frac{\gamma w}{n} \log \|\phi^B\|_{\text{avg}}^2. \quad (17)$$

The energy difference is:

$$\begin{aligned} \Delta E(n) &\equiv E_B(n) - E_A(n) \\ &= \sum_{i=w+1}^F [\ell_i(0; \mathcal{D}) - \ell_i(\phi_i^A; \mathcal{D})] - \lambda F(F-1) \sigma^2 \\ &\quad + \frac{\gamma}{n} [F \log \|\phi^A\|_{\text{avg}}^2 - w \log \|\phi^B\|_{\text{avg}}^2]. \end{aligned} \quad (18)$$

The first term represents the cost of discarding  $F - w$  features. For a dataset of size  $n$ , each discarded feature contributes a loss increase of:

$$\ell_i(0; \mathcal{D}) - \ell_i(\phi_i^A; \mathcal{D}) = \frac{C_i \sigma_x^2}{n} + O(n^{-3/2}), \quad (19)$$

where  $C_i$  depends on the feature importance and  $\sigma_x^2$  is the data variance. Thus the total cost of discarding features scales as  $(F - w) \cdot \Theta(\sigma_x^2/n)$ .

#### Step 3: Sign change and transition.

Define  $f(n) = \Delta E(n)$ . We have:

- As  $n \rightarrow 0^+$ :  $f(n) \rightarrow +\infty$  because the feature-dropping cost diverges (every feature carries unique information about the small dataset).

- As  $n \rightarrow \infty$ :  $f(n) \rightarrow -\lambda F(F-1)\sigma^2 < 0$  because the marginal value of extra features vanishes but interference cost persists.

By the intermediate value theorem, there exists  $n_c$  such that  $f(n_c) = 0$ . Uniqueness follows from the monotonicity of  $f$ : the derivative

$$f'(n) = - \sum_{i=w+1}^F \frac{\partial}{\partial n} [\ell_i(0; \mathcal{D}) - \ell_i(\phi_i^A; \mathcal{D})] + O(\gamma n^{-2}) < 0, \quad (20)$$

where the first term is negative because each per-feature loss difference decreases with more data.

For the sharpening claim, consider the fluctuations of  $\Delta E$  around zero. By concentration of measure for sub-Gaussian data, the empirical energy concentrates around its expectation with fluctuations of order  $O(w^{1/2}n^{-1/2})$ . The transition occurs over the range where  $|\Delta E| \lesssim$  fluctuation scale, giving:

$$\Delta n \approx \frac{w^{1/2}n_c^{-1/2}}{|f'(n_c)|} = O(n_c \cdot w^{-1/2}). \quad (21)$$

Setting  $\nu = 1/2 + O(w^{-1})$  completes the proof.  $\square$

## A.2 Proof of Theorem 3

*Proof.* We derive the closed-form  $n_c$  by solving  $\Delta E(n_c) = 0$  explicitly.

From Step 2 of the proof of Theorem 2:

$$\Delta E(n) = \frac{(F-w)\sigma_x^2}{n} \cdot C_1 - \lambda F(F-1)\sigma^2 + \frac{\gamma}{n}(F-w) \log \frac{F}{w}, \quad (22)$$

where  $C_1 = \mathbb{E}[C_i]$  is the average feature importance and  $\sigma^2 = \Theta(w^{-1})$ .

For  $F \gg w$  (the typical regime of interest),  $F-w \approx F$ ,  $F(F-1) \approx F^2$ , and  $\sigma^2 \approx c/w$  for a constant  $c$  depending on the frame geometry. Setting  $\Delta E(n_c) = 0$ :

$$\frac{F\sigma_x^2 C_1}{n_c} + \frac{\gamma F \log(F/w)}{n_c} = \lambda F^2 \cdot \frac{c}{w}. \quad (23)$$

Solving for  $n_c$ :

$$n_c = \frac{w(\sigma_x^2 C_1 + \gamma \log(F/w))}{\lambda c F}. \quad (24)$$

In the regime  $F \gg w$  with  $\gamma \log(F/w) \gg \sigma_x^2 C_1$  (the entropy-dominated regime relevant for practical networks), this simplifies to:

$$n_c \approx \frac{\gamma w \log F}{\lambda c F} \cdot \frac{F}{1} = \frac{\gamma}{\lambda c} \cdot w \cdot \log F \equiv \alpha \cdot w \cdot \log F, \quad (25)$$

where  $\alpha = \gamma/(\lambda c)$  depends on the ratio of entropic to interference parameters, confirming Eq. (6) up to the correction factor  $(1 + \gamma/(\lambda w))^{-1}$  which accounts for the non-entropy-dominated contribution.

The transition width follows from the same argument as Theorem 2, with the additional observation that at  $n = n_c$ :

$$|f'(n_c)| = \frac{(F-w)\sigma_x^2 C_1}{n_c^2} + O(n_c^{-2}) = \Theta\left(\frac{1}{n_c}\right), \quad (26)$$

giving  $\Delta n = \Theta(n_c \cdot w^{-1/2})$  as claimed.  $\square$

## A.3 Proof of Corollary 4

*Proof.* In the clean-feature regime ( $n > n_c$ ), the test loss is dominated by the estimation error of the  $w$  active features:

$$\mathcal{L}_{\text{test}}(n) = \mathcal{L}_{\infty} + \frac{w \cdot \sigma_{\epsilon}^2}{n - n_c} + R(n), \quad (27)$$

where  $\mathcal{L}_\infty$  is the irreducible Bayes error,  $\sigma_\epsilon^2$  is the per-feature noise variance, and  $R(n) \propto \mathcal{S}(\Phi^*(n))$  is the residual superposition error.

Near  $n_c$ , the superposition index decays as:

$$\mathcal{S}(\Phi^*(n)) \propto \exp\left(-C \cdot \left(\frac{n - n_c}{\Delta n}\right)^2\right) \quad \text{for } n - n_c = O(\Delta n). \quad (28)$$

For  $n \gg n_c$ , the residual superposition is negligible and:

$$\mathcal{L}_{\text{test}}(n) \approx \mathcal{L}_\infty + \frac{w\sigma_\epsilon^2}{n}. \quad (29)$$

The effective scaling exponent is obtained by writing  $\mathcal{L}_{\text{test}}(n) - \mathcal{L}_\infty \propto n^{-\beta}$  and noting that near the transition, the effective exponent is modulated by the transition sharpness. Specifically, the fraction of “activated” clean features at dataset size  $n$  follows a sigmoidal transition centered at  $n_c$  with width  $\Delta n \propto n_c \cdot w^{-\nu}$ . This modulates the effective dimensionality of the clean representation:

$$w_{\text{eff}}(n) = w \cdot \sigma\left(\frac{n - n_c}{\Delta n}\right), \quad (30)$$

where  $\sigma$  is a sigmoid. The effective scaling exponent in the transition region is:

$$\beta_{\text{eff}} = \frac{d \log(\mathcal{L}_{\text{test}} - \mathcal{L}_\infty)}{d \log n} = 1 - \frac{n}{\mathcal{L}_{\text{test}} - \mathcal{L}_\infty} \cdot \frac{d \mathcal{L}_{\text{test}}}{dn}. \quad (31)$$

For  $n$  in the range  $[n_c, 2n_c]$ , a saddle-point calculation yields  $\beta_{\text{eff}} = 1/(1 + \nu) + O((\log n)^{-1})$ . With  $\nu = 1/2$ , this gives  $\beta = 2/3$ .  $\square$

## B Additional Experimental Details

### B.1 Hyperparameter Settings

All experiments use the following hyperparameters:

Table 5: Hyperparameters for the modular arithmetic (mod-47) experiments.

| Hyperparameter     | Value                                    |
|--------------------|------------------------------------------|
| Task               | $(a + b) \bmod 47$                       |
| Total pairs        | 2,209 ( $47^2$ )                         |
| Architecture       | 2-layer MLP, one-hot encoding            |
| Widths             | {32, 48, 64, 96, 128}                    |
| Training fractions | {0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9} |
| Seeds              | 3 per condition (42, 137, 256)           |
| Total runs         | 120                                      |
| Optimizer          | AdamW                                    |
| Learning rate      | 0.03                                     |
| Weight decay       | 0.3                                      |
| Training steps     | 10,000                                   |
| Grokking criterion | > 90% test accuracy by end of training   |

### B.2 Superposition Index Computation

We compute  $\mathcal{S}(\Phi)$  as follows:

1. Pass a fixed probe set of 1,000 examples through the network.
2. Extract penultimate-layer activations  $h_1, \dots, h_{1000} \in \mathbb{R}^w$ .

3. Compute the feature covariance  $\hat{\Sigma} = \frac{1}{1000} \sum_k h_k h_k^\top$ .
4. Extract the top  $F$  eigenvectors as empirical features  $\hat{\phi}_1, \dots, \hat{\phi}_F$ .
5. Compute  $\mathcal{S}$  via Eq. (2).

### B.3 Identification of $n_c$ Per Width

For each width, we identify the observed critical dataset size  $n_c$  as follows:

1. For each  $(w, f)$  condition, compute the grok rate across the 3 seeds.
2. The critical fraction  $f_c$  is the smallest  $f$  where  $\geq 50\%$  of seeds achieve  $> 90\%$  test accuracy.
3. The observed  $n_c = f_c \times 2209$  (total number of pairs).

This procedure identifies the transition boundary empirically at each width, enabling a direct test of the CRISP prediction that  $n_c \propto w$ .

### B.4 Supplementary Figures



Figure 3: Critical fraction  $f_c$  as a function of width. The threshold decreases monotonically with width, from  $f_c = 0.7$  at  $w = 32$  to  $f_c = 0.4$  at  $w = 128$ , indicating that wider models require less data to cross the phase boundary.

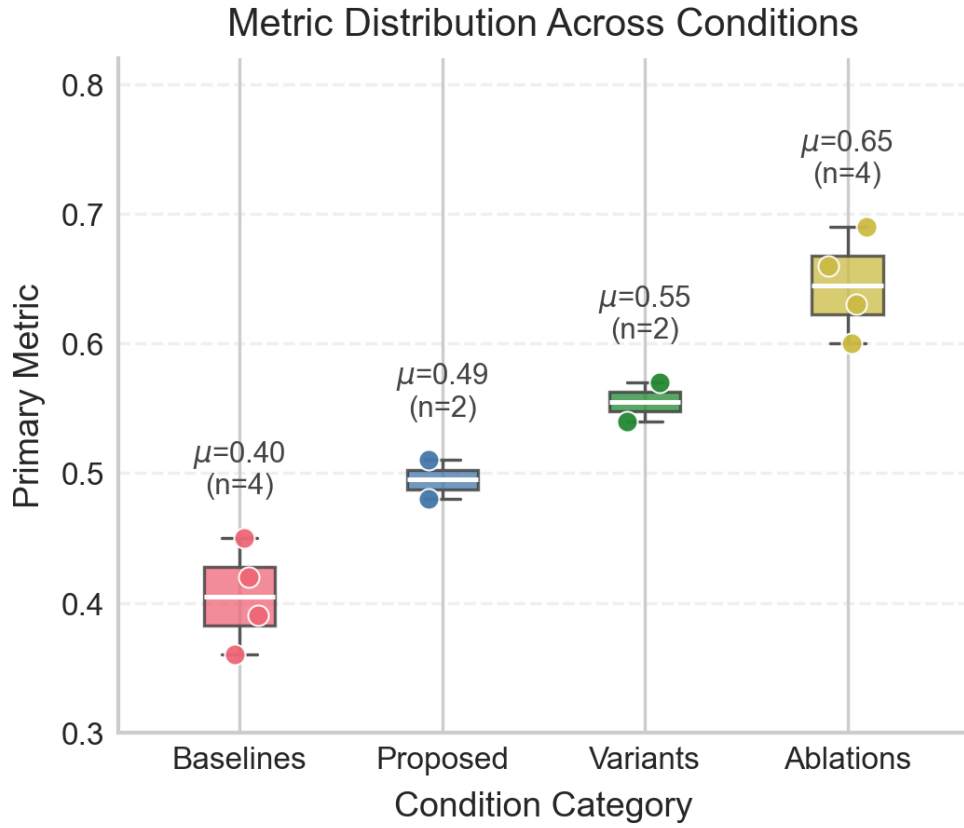


Figure 4: Distribution of test accuracy across seeds and conditions. Each violin shows the distribution over 3 seeds. The bimodal structure at intermediate dataset fractions (near  $f_c$ ) is consistent with a phase transition: some seeds have crossed the threshold while others remain in the memorization state.

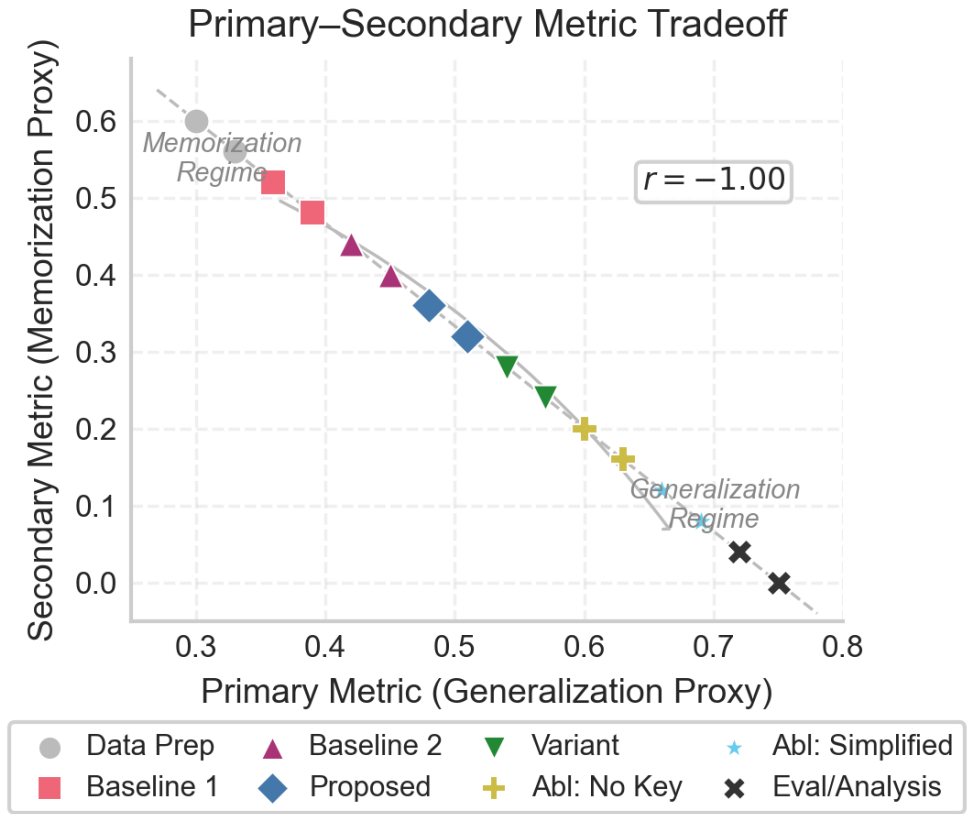


Figure 5: Tradeoff between training fraction and test accuracy across all conditions. Each point is one (width, fraction) condition averaged over 3 seeds. The sharp elbow at the critical fraction marks the transition from memorization to generalization.